

Graeme Caiger

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Experienced data professional proficient in SQL, Python, PySpark and C++, with proven ability to develop robust data pipelines, both independently and collaboratively (Git). Advanced Excel skills, including VBA and Power Query, as well as strong capability with Power BI and DAX.

Skilled in data modelling, cleansing and integration, with experience using web APIs, deriving clear, actionable insights, and communicating to non-technical audiences. Able to apply machine learning pragmatically to a range of problems, from simple regression to fine-tuning modern transformer models. Known for approachability, attention to detail, calm delivery under pressure, and a commitment to continuous learning.

Experience

Metropolitan Police DDaT: Junior Data Engineer *(Jan 2026 – May 2026, Extended Secondment)*

- Delivered multiple end-to-end data pipelines as part of the migration to Azure, using Databricks, ADLS Gen2, Alteryx, Azure Data Factory, SQL, Power Automate and Python.
- Leveraged PySpark to optimise data processing workflows and conduct data quality analysis, identifying and investigating key cross-system discrepancies and recommending remediations to stakeholders.
- Built PySpark libraries/snippets used by colleagues for deterministic testing, QA and IO for Excel and Parquet files.
- Performed CONNECT Data Reconciliation Process and learned about the transformations from Mirror-POLE to analysis ready datasets.

Metropolitan Police DDaT: Data Quality Assistant *(Oct 2024 – Present)*

- Developed SharePoint-integrated Power BI dashboards delivering regular performance reporting for National Data Quality Improvement Service work.
- Owned end-to-end data modelling, transformation and validation for ADR351 (Drugs), using SQL, DAX, VBA and Power Query to ensure accuracy and integrity; introduced automation and usability improvements that increased team output by approximately 300% and enabled tracking of key performance indicators and data quality metrics.
- Independently investigated and led the resolution of complex data quality issues using descriptive, diagnostic, predictive and prescriptive analysis: quantified the scale of issues, identified root causes, anticipated downstream impacts and future risk, and implemented targeted fixes. For example, I identified a systemic date accuracy edge case affecting aggregation, analysed impacted records, and introduced code and process changes to prevent recurrence and ensure robust handling.
- Built effective working relationships across internal teams and external stakeholders such as the Home Office ensuring alignment with national standards and requirements.
- Led workshops, training sessions and presentations to improve data quality practices and technical literacy. For example, presenting on the advanced features of CONNECT search, Power Query and drug code abstractions.
- Developed rule-based (decision tree) and transformer-based (ALBERT) NLP classification pipelines to detect data quality issues and prioritise high-risk records for manual review, using appropriate evaluation metrics to assess model performance and effectiveness.
- Trusted to independently carry out detailed data quality checks and amendments on CONNECT, applying strong data governance and ethical standards by ensuring information is accurate, proportionate, and only accessed or retained for a clear policing purpose.

Sports Direct, Sales Assistant *(Nov 2023 – Oct 2024)*

- Worked in a high-tempo environment balancing competing priorities and delivering on performance metrics.
- Empathetically engaging with each customers unique needs and matching them with products that best suit them.

Research Student Placement: RHUL, Physics Department *(Jun 2021 – Aug 2021)*

- Review and amendment of the serialisation and storage approach for data intensive ultra-low temperature experiments. Involved writing code with Python, Hadoop and LabVIEW.

Education

BSc (Hons) Physics [2:2]: *Royal Holloway University of London (RHUL) (Sept 2018 – Jul 2021)*

- Mathematical Modelling in MATLAB and Mathematica. GUI programming, C++, Python for machine learning and data analysis (Scikit-learn, Pandas, Matplotlib).
- Scientific report writing (LaTeX). An emphasis on independent data analysis and problem-solving skills, stringent deadline management, personal organisation, communication skills and tech literacy including the practical application of supervised and unsupervised machine learning.
- *Final year project: 'Modelling heat transfer through a metallic sinter using finite element analysis (COMSOL)'.*
- Specialisation: Machine Learning (Python), C++, Advanced Classical Physics (Lagrangian Mechanics, Variational Calculus), Superconductivity and Magnetism, Planetary Geology and Routes to Chaos (Dynamical Systems).

Level 3 Extended Diploma Games Design [Triple Distinction]: *North Kent College (Aug 2015 – Jun 2017)*

- Exposure to project lifecycle and management frameworks (Agile and Scrum methodologies) and a range of other skills such as sketching, 3D modelling, programming (C#), rigging, animation and Unity engine.

A Level Maths & Physics [B, D]: *Bexleyheath Academy (Aug 2015 – Jun 2017)*

AS Level Biology & Computing [C, D]: *Longfield Academy (May 2023), Gravesend Grammar (Sept 2014 – Aug 2015)*

Level 3 Speech & Drama [Grade 7]: *Trinity College London (2015)*

5 GCSE (including Maths [A] & English [A*]) [1A*, 2A, 2B, 1D]: *Rochester Independent College (2009 – 2014)*

Projects

- Gym fitness tracker integrating Notion API with Power Query and Power BI to produce a personal performance dashboard. Enabling the visualisation of progress on training volume over time and measures like BMI.
- Neural network (MLP) from first principles using NumPy. Model performance visualised in Matplotlib.
- Lecture transcription project using recursive folder exploration and FFmpeg to prepare media, run services (Python) and download data using the Falcon AI API.
- Air quality monitor (sensor datasheets, git C#, Node Red output to a web dashboard.), Beautiful Soup powered web scraper [HTML, CSS, Python]. Antiques BI (Excel Pivot table Dashboards, Python), Stock Market Simulator (Java), Gibbs Phenomenon simulation app (GUI powered by Python, Tkinter and matplotlib)

Additional Qualifications & Other Skills

- DataCamp: Data Analyst in Databricks, Associate Data analyst and Associate Data Engineer career tracks. Also courses in Power BI, Snowflake SQL, Data Communication, Data Ethics, Database Design, Cloud Computing, Machine Learning, Regex, Shell, Power Query, M, DAX and Excel.
- Advanced Office (Word, PowerPoint, Excel, Acrobat, etc)
- College of Policing E-Learning on Information management and security.
- Full clean UK driving licence.
- Saint John Ambulance Volunteer Vaccinator (Including basic life support training).
- PADI Advanced Open Water Diver.

Extra-Curricula

South London Makerspace/Hackerspace

- Community STEM outreach through South London Maker Festival. Arduino/Pi/Combat Robotics, Linux, CNC etc.

Lewisham Underwater Hockey [Club Secretary & Coach]

- Reviewing and analysing match recordings, creating individualised development plans and formulating new strategy.

Duke of Edinburgh [Bronze]

- Volunteering in the local community assisting elderly residents with everyday tasks and personal care. Independent hiking expedition across the English countryside.